NAGIZAAZ SHAIK

GENERATIVE AI ENGINEER · AGENTIC AI · RAG PIPELINES · AWS · FINTECH

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Generative AI engineer for fintech: five years in data and ML engineering, the last two building production LLM systems and AWS ML platforms for regulated credit unions, including RAG over regulatory documents and a governed LLM gateway. Doctoral researcher in AI-agent security; crewAI contributor.

SKILLS

Generative AI: Claude, OpenAI, Hugging Face, Qwen2.5, LoRA / PEFT, prompt engineering, structured outputs, vLLM, Ollama
Agentic AI: LangGraph, CrewAI, MCP, tool calling, multi-agent orchestration, agent memory, guardrails
RAG Pipelines: LangChain, embeddings, Pinecone, FAISS, OpenSearch, BM25 hybrid search, reranking, HyDE, corrective RAG
AWS: Bedrock, SageMaker (endpoints, Pipelines, Model Monitor), Lambda, S3, Glue, Redshift, EKS, IAM, CloudWatch
Eval & Safety: RAGAS, LLM-as-judge, held-out tests, OWASP LLM Top 10 red-teaming, prompt-injection tests, PII redaction
MLOps & Data: Databricks, MLflow, Airflow, Kafka, FastAPI, Docker, Kubernetes, Terraform, ArgoCD, GitHub Actions, n8n
Fintech & Languages: credit-risk modeling, ECOA fair lending, NCUA compliance, SHAP | Python, SQL, Go

EXPERIENCE

ML Platform Engineer · Contract | Wright-Patt Credit Union ($9B assets), Fairborn, OH *Jul 2025 – Jun 2026*

– Built the credit union's internal GenAI gateway (Claude, OpenAI, Ollama) with evaluation guardrails, cost controls, and RAGAS grounding checks: **one governed path** for the 12–15-person IT team to ship GenAI features.
– Founded the AWS ML platform: SageMaker endpoints and Pipelines, MLflow registry with gated promotion and automated rollback, per-environment VPC isolation; **deployment failures down 85%** in 60 days.
– Owned ML data security (AES-256 at rest and in transit, least-privilege IAM, audit logging); passed **two consecutive NCUA audits with zero findings**.
– Codified the infrastructure in Terraform with GitOps on ArgoCD, making every environment change reproducible and peer-reviewed.
– Mentored two engineers into ML and platform ownership through weekly architecture and code reviews.

Machine Learning Engineer · Contract | Day Air Credit Union ($830M assets), Kettering, OH *Jul 2024 – Jun 2025*

– Built a RAG pipeline over 500+ internal policy and NCUA documents (LangChain, vector search, Claude); compliance research went **from ~2 hours to under 20 minutes** per question.
– Fine-tuned a 14-class BERT intent classifier to fix member-support misrouting, served on FastAPI at **sub-200 ms p99**.
– Built the first-pass consumer-loan underwriting model (XGBoost on SageMaker) as the first dedicated ML engineer, keeping approval rates within risk thresholds.
– Replaced a manual multi-hour retraining cycle with a weekly Airflow pipeline: rolling-window retraining, holdout evaluation, and AUC / precision / calibration promotion gates.
– Set up SageMaker Model Monitor drift detection with custom baselines; it caught **two data-drift events** before they reached credit decisions.

Senior Data Engineer | Cognizant (financial-services client), Hyderabad, India *Jul 2020 – Jun 2023*

– Led the Oracle 11g to Amazon Redshift migration for five business units, redesigning the schema for columnar storage and rebuilding 20+ stored procedures as Airflow DAGs; **query runtime down 60%**.
– Built 15 production ETL/ELT pipelines with a multi-stage validation layer that caught a silent upstream corruption affecting three months of financial reports.
– Promoted from Associate to Senior in two years; led code reviews for three engineers and was primary client contact on two accounts.

Python Intern | Uber, Hyderabad, India *Jan 2019 – Oct 2019*

– Automated a 4-hour weekly manual ETL job as a scheduled Python job with anomaly checks on 7-day baselines; earned a return-internship offer.

RESEARCH

Doctoral Researcher, Applied AI | Belhaven University, Jackson, MS *Aug 2026 – Present*

- Traced an AI-agent sandbox bypass into 25 open-source projects that copied the vulnerable interpreter; **23 were still unpatched**. Reporter credit on GitHub advisory GHSA-m5v3-6ccr-cfgx (working paper *When Vendoring Defeats the Patch*).
- Released guard-gap-detector, a read-only static checker with a synthetic corpus, so the finding is reproducible without exposing unpatched projects.
- Showed that a planted label pushes a LoRA-tuned financial classifier to the attacker's label on **up to 92%** of items (five markets, three seeds, controls); a standard confidence gate missed about half of one attack's flips (working paper *Fragile in Different Ways*).

PROJECTS

Finance Agent Crew | Python, asyncio, Claude, FastAPI github.com/shaikn6/finance-agent-crew

A ticker goes in, a sourced equity-research brief comes out: concurrent SEC EDGAR, market, and news collection feeds parallel fundamental, sentiment, and risk agents, and a writer agent returns a typed BUY / HOLD / SELL report; a failed data source is recorded, not fatal.

LLM Gateway | Python, FastAPI, Redis, Docker github.com/shaikn6/llm-gateway

One OpenAI-compatible API over Anthropic, OpenAI, and Ollama with per-tenant caching, rate limiting, A/B routing, cost tracking, and audit logs; contract tests catch breaking provider SDK changes in CI.

Financial Text Classification with LoRA | PyTorch, PEFT, Qwen2.5, FinBERT github.com/shaikn6/fin-lora

A LoRA-tuned Qwen2.5-0.5B trained on tweets scored only 59.0% on news; retrained on news, tweets, and headlines, it reached **84.7%** vs 71.7% for TF-IDF and 73.6% for FinBERT. A confidence cascade that sends only uncertain items to the LLM cut average latency **63%** for 0.4 points of accuracy.

lethe: Agent Memory Server | Go, Python, LangChain, MCP github.com/shaikn6/lethe

Learns from feedback which memories to return to an agent. The first learner lost to a hand-set rule (0.611 vs 0.641 evidence in the top 10) because new memories were never tried; forced exploration fixed it: **0.707** on held-out LoCoMo conversations, at 7,119 requests/s.

LLM Safety Auditor | Python, sentence-transformers, FastAPI github.com/shaikn6/llm-safety-auditor

Red-teaming harness that scores **250+ adversarial attacks** against the OWASP LLM Top 10, with a 6-layer detector and PDF audit reports.

Open Source | 15 merged PRs: crewAI (60K stars, incl. a fix for a bug I reported), Ray (44K), MLflow (28K) github.com/shaikn6

EDUCATION & CERTIFICATIONS

Belhaven University — Doctor of Business Administration (DBA), Applied AI · GPA 4.0 / 4.0 Aug 2026 – Present

University of Dayton — M.S. Computer Science · GPA 3.93 / 4.0 Aug 2023 – May 2025

CMR Engineering College — B.Tech. Computer Science and Engineering · GPA 8.9 / 10 Aug 2016 – May 2020

AWS Certified Machine Learning – Specialty • AWS Certified Data Engineer – Associate • GCP Professional Data Engineer